

A Reinforcement Learning Architecture for Spatio-Temporal Reallocation in Real-Time Operations with Delayed Effects

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Abstract. Terminal assignments can become inefficient as yard and gate congestion changes. We study an architecture that revisits the yard block and arrival time of an announced truck job. Separate proposal and acceptance networks learn pair-centred cost scores from simulation branches with delayed consequences; a central procedure checks mutual acceptance and simulated resource feasibility. Evaluation uses fixed networks without online counterfactual simulation. In the original single continuous synthetic run the joint policy reduced modelled terminal cost by 14.72% over the middle 28 days against 13.44% for time-only adjustment. In a registered evaluation of 20 independent months, time-only adjustment was cheaper than no reallocation in 16 months (median 12.7%), and in all 14 months free of a simulator crane deadlock, at the same completion rate and with a positive 97.5% interval; the joint policy was cheaper in only 4 months, so spatial adjustment did not replicate. Appointment quotas are unbounded and external waiting and rescheduling costs are excluded. The architecture is positioned as a decision-support component for terminal staff and carrier dispatchers; neither deployment feasibility nor user benefits have been established.

Keywords: Container terminal, Truck appointment system, Yard block reallocation, Counterfactual cost learning, Reinforcement learning

1 Introduction

Terminal automation must turn information into coordinated operational decisions [1]. Appointment systems address arrival windows and carrier-terminal coordination [2, 3]; yard scheduling addresses placement, sequencing, and interference [4–7]. Reinforcement learning also supports multi-block crane dispatching [8].

We ask whether an announced job’s block or arrival time should change as congestion develops. Effects on later trucks, cranes, and vessels make immediate movement cost insufficient. Difference rewards and counterfactual baselines motivate matched action comparisons [9, 10]. Our separate proposal and acceptance networks precede central commitment; Fig. 1 distinguishes offline learning from online decisions.

The contributions are the assignment architecture, its pairwise learning signal, and an evaluation protocol separating action-range effects from within-run dependence. Historical results are a synthetic proof of concept, not evidence of consistent benefits or component necessity. Sect. 3.5 describes the intended decision-support users; field and human benefits remain untested.

2 Related Work

Appointment models redistribute arrivals [2] and collaborative scheduling represents carrier and terminal interests [3], while empirical time-of-day models motivate nonuniform arrival profiles [11]. Appointment quotas trade truck turn time against crane utilisation [12], so our unbounded-quota experiments address adjustment potential under stated assumptions.

Yard research studies crane sequencing [4], interference [5], block length and crane deployment [6], and demand-dependent dispatching performance [7]. Recent reinforcement learning addresses multi-block crane dispatching [8]. We instead revise the assignment supplied to the execution layer, while keeping that layer’s dispatching rule fixed within a comparison.

Difference rewards and counterfactual multi-agent learning isolate an actor’s contribution to a shared outcome [9, 10]. Here the baseline comes from a simulation branch and the target is a finite-horizon cost difference, so unlike temporal-difference targets in deep Q-learning [13] it does not bootstrap from the network’s own future estimates. Small multilayer perceptrons [14] map observations and candidate features to scalar scores; approximation capacity alone does not guarantee correct candidate ordering, and Sect. 3.3 states the remaining gap.

3 Proposed Architecture

3.1 Notation, Actions, and Policies

Every $\delta = 60$ s, eligible announced jobs are reviewed once. Job i has assignment $z_i = (b_i, \tau_i)$ and legal actions $a_i = (b'_i, \Delta_i) \in \mathcal{A}_i(t)$, producing $z_i(a_i) = (b'_i, \tau_i + \Delta_i)$. KEEP is $a_i^0 = (b_i, 0)$. Full selects among spatial and temporal alternatives; each changed candidate alters one axis: block or arrival delay ($\Delta_i > 0$). Outbound jobs retain their assigned block. Let n_i be announcement time, g_i gate-in time, and r_i the prior-review indicator. With $W = 1800$ s and $w_i = \min(W, \tau_i - n_i)$, eligibility is

$$\mathcal{E}_t = \{i \mid w_i - \delta < \tau_i - t \leq w_i, \tau_i > t, g_i > t, r_i = 0\}. \quad (1)$$

Review occurs within one decision interval after entering this window, approximately W before arrival when notice is sufficiently early. Jobs announced at or after arrival are ineligible; the lead-time model assigns 14.1% of demand to that category. This review window is an algorithmic choice, not a validated carrier notification allowance. The proposal policy scores each candidate from the block

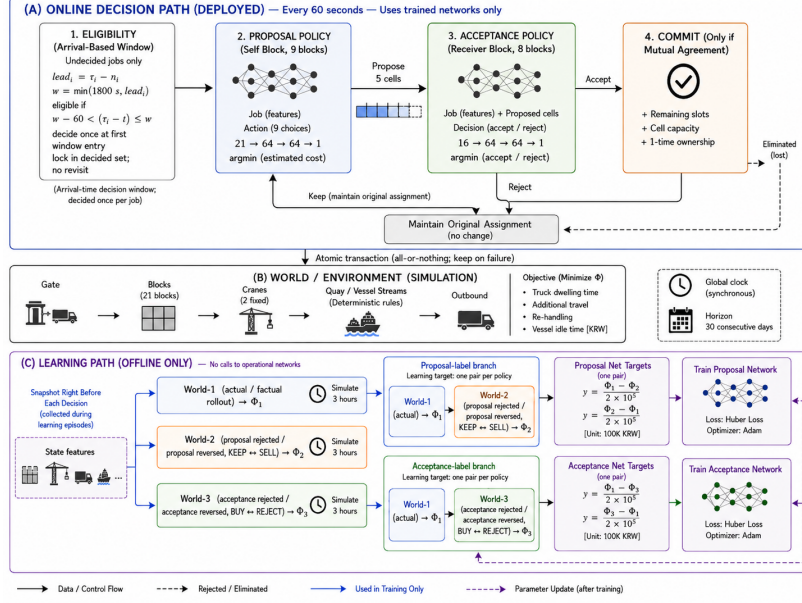


Fig. 1: (a) Every 60 s, the online path checks eligibility, proposal, acceptance, and commitment. (b) The simulator links gate, blocks, cranes, and vessels to cost Φ . (c) Offline learning compares up to three worlds from the same state and random stream over 3h to label the two networks.

state and job-action features o_i^P and selects the lowest score, while the acceptance policy compares acceptance with rejection from the proposal message and the destination state o_i^A :

$$\hat{a}_i = \arg \min_{a \in \mathcal{A}_i(t)} \hat{c}_\theta^P(o_i^P, a), \quad p_i = \mathbf{1}[\hat{a}_i \neq a_i^0],$$

$$q_i = \mathbf{1}[\hat{c}_\psi^A(o_i^A, \hat{a}_i, \text{acc}) \leq \hat{c}_\psi^A(o_i^A, \hat{a}_i, \text{rej})]. \quad (2)$$

If the target resource has no remaining capacity, the request is rejected before the network is called. The mutually accepted candidates form $\mathcal{J}_t = \{i \in \mathcal{E}_t \mid p_i q_i = 1\}$.

3.2 Central Commitment

The central procedure creates no new candidates. It considers only $\mathcal{J}_t$, ordered by the acceptance policy's predicted score with lexicographic tie-breaking. The pair-centred scores are not calibrated across jobs (Sect. 3.4). Their order is deterministic, but can change which proposals obtain a contested resource; it is therefore an algorithmic priority, not a validated ordering of monetary benefits. Its effect must be separated from the acceptance veto. With $x_i \in \{0, 1\}$ indicating commitment, $F_b(t)$ the free slots in block b , and $K_k(t)$ the remaining quota of time slot k , it enforces

$$\begin{aligned}
x_i &\leq p_i q_i, & \sum_{i \in \mathcal{J}_t} x_i \mathbf{1}[b'_i = b] &\leq F_b(t), \\
\sum_{i \in \mathcal{J}_t} x_i \mathbf{1}[k(i) = k] &\leq K_k(t), & \sum_{t \in \mathcal{T}} \mathbf{1}[i \in \mathcal{E}_t] &\leq 1.
\end{aligned} \tag{3}$$

Block-capacity checks are applied, but $K_k(t) = \infty$ in all reported experiments because terminal-specific booking quotas are unavailable. The candidate time range is an algorithmic setting, not a verified carrier availability window or notification allowance. Thus these checks establish feasibility only within the simulated rules, not the executability of a changed appointment in practice. All committed assignments are applied together at the end of the decision epoch, so an earlier commitment cannot change the state seen by another candidate in the same epoch.

3.3 Cost Networks

Both $\hat{c}_\theta^P$ and $\hat{c}_\psi^A$ are multilayer perceptrons with two 64-unit hidden layers and ReLU, with weights shared across blocks: $21 \rightarrow 64 \rightarrow 64 \rightarrow 1$ for proposal and $16 \rightarrow 64 \rightarrow 64 \rightarrow 1$ for acceptance, or fewer than 11,000 parameters in total. Inputs are order records and yard-state summaries only: current waiting, inventory, remaining crane work, the public expected arrival time, route difference, and announced volume near the target time; unrealised arrival and completion times and simultaneous responses from other accepting parties are excluded. The output is one pair-centred cost score per candidate, so the length of the candidate list may change without retraining. Pair centring preserves the ordering inside one pair, not every comparison in (2). For a fixed state and a common KEEP outcome $K = C(s, \text{KEEP})$, the ideal label of a changed candidate is

$$y(s, a) = \frac{C(s, a) - K}{2S}, \quad S = 100,000 \text{ KRW}. \tag{4}$$

Changed candidates preserve their true cost order only when the state, exogenous trajectory, continuation policy, and horizon share that same reference. The learned predictions need not preserve it. Moreover, KEEP's target $(K - C(s, a))/(2S)$ depends on the paired alternative, even when its input is unchanged. Ordering changed candidates and deciding against a learned KEEP score are therefore distinct validation questions. This conditional explanation does not establish measured ranking accuracy.

3.4 Counterfactual Cost Labels and Training

For each sampled decision we clone the state and the random stream just before branching and run up to three worlds for $H = 10,800$ s: $\phi_i^F = \Phi_H(\text{observed})$, $\phi_i^P = \Phi_H(\text{proposal reversed})$, and $\phi_i^A = \Phi_H(\text{acceptance reversed})$. The counterfactual branch reverses the policy's observed decision: if the proposal policy proposed a change the alternative keeps the assignment; if it kept the assignment,

the alternative forces a proposal selected by the current weights. Acceptance and feasibility still govern actual commitment. The acceptance alternative swaps acceptance and rejection. For each policy the two actions of the pair are centred on their mean and divided by KRW 100,000,

$$\bar{\phi}_i^P = \frac{\phi_i^F + \phi_i^P}{2}, \quad y_{i,F}^P = \frac{\phi_i^F - \bar{\phi}_i^P}{10^5}, \quad y_{i,P}^P = \frac{\phi_i^P - \bar{\phi}_i^P}{10^5}, \quad (5)$$

that is, $\pm$ half the pairwise cost difference; the same centring is applied to (ϕ_i^F, ϕ_i^A) . The reference point is therefore the pair itself rather than a fixed action. We sample at most 64 decisions per day, split them 80/20 into training and validation, and optimise with Adam (3×10^{-4}) and Huber loss ($\beta = 1.0$), which prevents a few very large counterfactual differences from dominating the fit [15]. Exploration decreases linearly from 0.50 to 0.05 over the 30 training iterations. States and alternatives change with the current policy; each iteration discards its labels after updating. This is finite-horizon Monte-Carlo cost learning with simulator-generated baselines, without target bootstrapping or a policy-gradient update. Training both networks for 30 iterations took about 18 hours of wall-clock time. During evaluation the weights are frozen, exploration is disabled, and no counterfactual branch is called, so network scoring is only part of the online path. Candidate construction, conflict resolution, and commitment must also be timed.

3.5 Intended Users and Operational Context

Terminal yard planners and operations staff are the intended users, with carrier dispatchers and drivers. Busan New Port waiting-time research [16], Korean terminal operating system (TOS) research [17], and the Busan Port Authority’s port community system with vehicle booking, documented by the World Bank and IAPH as a digitalization case [18], motivate this proposed application; integration is untested.

A planner would compare the original and recommended block/time against observed congestion and accept, modify, or reject; a carrier dispatcher would check vehicle availability and subsequent jobs and negotiate or decline; the driver would receive the agreed assignment and report execution difficulties. Operator review precedes carrier agreement and driver notification, and without agreement the original assignment stands. Human response deadlines are separate from the 60-second review interval.

Network acceptance is not human consent, and scores are neither calibrated savings nor confidence probabilities. The evaluated engine supplies no evidence about interfaces, usability, or human acceptance.

4 Experimental Environment and Evaluation Protocol

The synthetic setting draws on reported block counts [7], cranes [5], multi-week operation and handling times [6], and arrival variation [11]. Table 1 distinguishes

design choices from literature-informed values [19]; it does not reproduce a particular terminal. The generator’s 45% initial fill corrects the earlier 65% manuscript entry, not a new occupancy experiment. The cost attributed to day d has four terms. Truck costs follow the original requested day, while the other terms follow calendar time; consequently this is not a purely calendar-day cash-flow total. Here h_i is the dwell time of truck i , m the additional crane working time, R the number of rehandles, and T_v the policy-related idle time of vessel v :

$$\begin{aligned}\Phi_d &= C_{wait,d} + C_{move,d} + C_{rehandle,d} + C_{vessel,d}, \\ C_{wait} &= \sum_i 40,000 \{h_i + \max(0, h_i - 1)\}, \quad C_{move} = 60,000 m, \\ C_{rehandle} &= 2,000 R, \quad C_{vessel} = \sum_v 2.99 GT_v T_v.\end{aligned}\tag{6}$$

The base monetary scale is derived from the safe trucking freight rate [21], whereas the conversion to an hourly dwell-cost coefficient is a design choice.

Table 1: Simulation and learning settings. Design values, including literature-informed conversions, are not measurements at a specific terminal. Historical and revision evaluations are listed separately. Requested arrival times follow $f(t) = 0.38U(0, 24) + 0.62\sum_{m=1}^3 w_m \mathcal{N}(t; \mu_m, \sigma_m^2)$ with (μ_m, σ_m, w_m) equal to $(10, 1.5, .317)$, $(15, 2.5, .633)$, and $(21, 1, .05)$, time in hours.

Group	Setting	Value
Yard	Blocks, cranes per block	21, 2
	Slots per block, initial fill (design)	1,440, 45%
Time	Decision interval δ	60 s
	Review window W	1,800 s
	Counterfactual horizon H (design)	10,800 s
Demand	Announcement lead	median 1.27 h (14.1% ≤ 0)
	Daily levels (design)	3,500, 5,000, 7,500, 12,500, 15,000
	Draw probabilities (design)	0.30, 0.30, 0.25, 0.10, 0.05
	Arrival curve (design)	uniform base 38% + 3 Gaussians
Vessels	Classes (GT), calls/day, crane rate	50/100/150k, 3.3, 27.5 moves/h
Cost	Truck dwell (design)	KRW 40,000/h (twice beyond 1 h)
	Crane move, rehandle (design)	KRW 60,000/h, KRW 2,000/event
	Vessel idling	KRW 2.99 per GT-hour [20]
Learning	Networks (proposal, acceptance)	21/16 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 1
	Iterations, decisions per day	30, ≤ 64
	ε , optimiser	0.50 $\rightarrow$ 0.05, Adam 3×10^{-4}
Evaluation	Historical days, compared, seed	30, middle 28, 9,900,950
	Registered revision	20 months, 4 fixed-policy variants
	Policy at evaluation	frozen weights, $\varepsilon = 0$

Structural vessel time that the policy cannot change is recorded separately. Truck dwell is measured after gate entry; the objective does not include waiting outside the gate, carrier rescheduling charges, or disruption to subsequent trips. These omitted burdens can change the net benefit of a time adjustment. In the historical run, truck waiting contributes 96.4% of cost, making results especially sensitive to its coefficient and the excluded external burdens.

4.1 Historical Run and Independent Evaluation

Historical results use one continuous 30-day trajectory (seed 9,900,950), measured over days 2–29. Common exogenous streams support matched comparisons [22], but state carries across days. Daily wins are descriptive; day-level tests and day-resampled intervals are not independent-replication evidence. The first-iteration checkpoint already contains one update and is not an untrained model.

The revision fixes 20 seeds, starting at 20,000,000 with spacing 100,000. Each policy–seed run starts a fresh engine for 30 continuous days plus the existing two-hour closing interval. Arrival curves, daily-volume probabilities, truck attributes, and initial inventory are preserved and matched within seeds. Weights are fixed; exploration, updates, and training branches are disabled. This evaluates variation across environments for one model, not across training runs, and those weights were trained under the legacy admission contract, before the correction described next.

Legacy admission could omit requests lacking stock or reservations; the revision registers all requests and shares a fixed correction to selected vessel loading/discharging directions across policies, recording both input versions. Historical and revised costs are not pooled, and completed, unfinished and unbound work accompanies every cost. Outbound jobs retain their block, but synthetic targets may be rebound within it rather than representing fixed container identities.

No reallocation, Full, and Time-only comprise the original 60 runs; Block-only adds the same 20 seeds, totalling 80, under a separate 17 September registration made after one baseline month and partial results were available. All variants share the checkpoint, and action restrictions are not separately optimised policies.

For policy p and seed s , let $C_{p,s}$ be the cost over the 28 measurement days. The two original primary differences are

$$D_s^B = C_{\text{baseline},s} - C_{\text{Full},s}, \quad D_s^T = C_{\text{Time},s} - C_{\text{Full},s}. \quad (7)$$

Positive values favour Full. We resample the 20 paired months 20,000 times (seed 9,900,721), retaining within-month dependence: each draw takes 20 months with replacement and recomputes the mean difference, and the interval spans the central 97.5% of those means, so an interval excluding zero means the sign does not depend on which months were drawn. Each primary mean difference has a 97.5% percentile interval; secondary intervals are descriptive 95%. The

separate Block-only family uses seed 9,900,723 and 97.5% intervals for baseline minus Block-only and Block-only minus Full; two families do not provide one four-claim error guarantee. Per-seed costs, percentage savings, aggregate ratios, wins/ties/losses, and residuals require complete matched sets with no outcome-based sample changes; results are in Sect. 5.4. Daily records (volume, backlog, block queues, turn times, actions, costs) are descriptive; the 600 days are not independent replications.

5 Results and Evidence Status

5.1 Historical Cost Comparison

Table 2 preserves the legacy single-run results under the historical contract: KRW 10.00 billion without reallocation against 8.52 with Full over 28 days.

Table 2: Historical 28-day comparison. Positive reduction means lower recorded cost than no reallocation. Daily wins are descriptive, without independent-sample significance; dashes denote unreported values.

Policy	Actions	Reduction	Full cheaper (days)
Full	block+time	+14.72%	—
Time-only	time	+13.44%	—
Block-only	block	−1.31%	—
First-iteration model	block+time	+7.97%	18/28
Rule: least-loaded slot	time	+3.41%	—
Rule: least-loaded block+slot	block+time	+1.51%	20/28
Rule: least slack	block	+0.53%	20/28
Rule: first-come-first-served	block	+0.17%	22/28
Rule: shortest processing time	block	+0.05%	21/28
Rule: net gain	block	−0.48%	21/28
No reallocation	—	0.00%	26/28

Full’s additional reduction over Time-only is only 1.28 percentage points, and Block-only increases aggregate cost, so neither established a general need for spatial adjustment; Sect. 5.4 answers it. The least-loaded-slot rule saves 3.41% without learning, and historical Time-only beats it on just 8/28 days yet by about KRW 1.00 billion in aggregate: win frequency and magnitude must be read together.

5.2 Demand Level and Action Contributions

Four high-volume days contribute 90.5% of historical savings, with Time-only and Block-only at 1.344 and −0.131 billion KRW; the 0.258 billion residual is

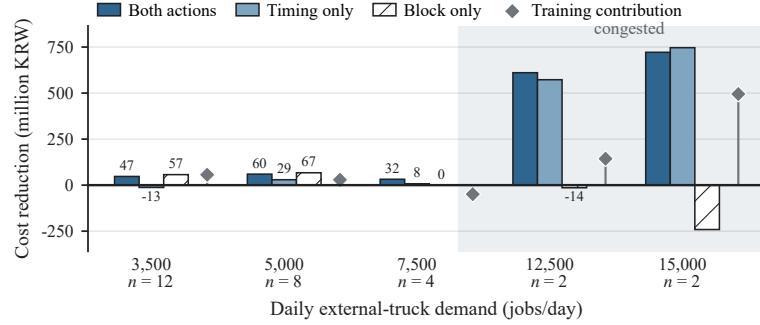


Fig. 2: Historical cost reduction by planned demand, in million KRW. Bars compare each restricted policy with no reallocation; the diamond compares Full with the first-iteration checkpoint. Shaded demand levels contain four dependent days, so totals are descriptive.

arithmetic, not a verified component effect. Block-only saves at 3,500 and 5,000 trucks but loses at 15,000, so spatial benefits were already conditional. Volume alone does not establish congestion when backlog persists across days.

An existing sweep (2, 6, and 14 high-volume days out of 28, seed 9,900,980) puts Time-only at 67%, 79%, and 81% of Full’s savings; it validates no other peak width, cost, or horizon.

5.3 Learning Diagnostics

Over the historical 30 iterations (4,671 counterfactual worlds), proposal training and validation losses fell from (0.079, 0.268) to (0.029, 0.119) and acceptance losses from (0.171, 0.315) to (0.027, 0.136); pair-fit diagnostics do not prove global ranking. The first-iteration checkpoint costs KRW 9.20 billion and later updates save another KRW 0.675 billion, 94.6% of it on four high-volume days.

5.4 Independent Months

Table 3 reports the registered comparisons. Time-only was cheaper than no reallocation in 16 months (median saving 12.7%), whereas Full was cheaper in 4 and Block-only in 3; every pre-registered mean-difference interval includes zero. Nine runs in six seeds, two of them no-reallocation runs, ended in a simulator deadlock: two idle cranes at a block end sat within each other’s safety gap with all pending work between them, and the feasible move-aside reposition the generator emits was discarded once higher-scoring blocked jobs filled the candidate budget, idling the block for as long as 23.5 days. These runs, not the policies, drive the means; the registered table retains them. On the 14 seeds where no arm stalled for 24 h or more, a mechanical criterion applied to every policy, Time-only is cheaper in 14/14 months while Full and Block-only are significantly more expensive. That

subset is not neutral: the six removed seeds average 7.2% more requested trucks (211,917 against 197,607), so the filter discards the busiest months.

Three checks bound the Time-only saving. It is not an artefact of unfinished work: on those seeds the policy completes 99.99% of demand against the baseline’s 100.00%, at KRW 53,520 per completed truck against 61,939, cheaper in 14/14 months. The 30-day window agrees (14/14, median 14.19%). And the omitted external cost bounds it: over 1.60 million changed appointments, a rescheduling charge above about KRW 14,600 each, or KRW 17,700 per deferred hour at a mean deferral of 49 minutes, would erase it.

The spatial result has a mechanism: enabling block changes makes the policy give up 305,093 temporal adjustments for 393,404 spatial ones, turning a KRW 23.3 billion saving into a KRW 37.8 billion loss on the same seeds, and leaves targets unbound in most seeds. Spatial adjustment therefore does not replicate for this trained model and simulator revision. Per-seed costs accompany the response.

Table 3: Independent 28-day comparisons. Mean saving in billion KRW with percentile intervals from 20,000 month resamples (97.5% for primary and added-family rows, 95% otherwise); wins count months where the second policy is cheaper. Lower rows keep only the 14 seeds in which no arm stalled for 24 h or more, and use a separate resampling seed.

Set	Comparison	Mean saving	Interval	Wins
20 months	Baseline vs Full	+2.39	[−11.73, +26.39]	4/20
	Time-only vs Full	+15.02	[−4.80, +41.91]	7/20
	Baseline vs Time-only	−12.63	[−42.37, +16.95]	16/20
	Baseline vs Block-only	−12.06	[−29.28, +14.04]	3/20
	Block-only vs Full	+14.46	[+7.81, +22.58]	19/20
Deadlock-free 14 seeds	Baseline vs Time-only	+1.66	[+1.24, +2.07]	14/14
	Baseline vs Full	−2.70	[−5.32, −0.36]	3/14
	Baseline vs Block-only	−17.06	[−25.49, −9.44]	2/14

5.5 Outstanding Evidence and Reporting Boundaries

The repair, ranking feasible candidates ahead of blocked ones, is implemented and tested; it alters the candidate set whenever the budget binds, so the re-run needs fresh pre-registration. The deadlock-free rows are a validity filter, not a registered analysis, and they drop the busiest months, so they may flatter a policy whose savings concentrate in congestion. The evaluation also varies one axis: one set of trained weights, produced under a different admission contract, applied to 20 environments, so training-seed stability is untested. The four policies do not isolate the acceptance network: veto removal retains score-based ordering, and a separate ablation must distinguish these functions.

Candidate ranking remains unvalidated: KEEP and every legal action must be compared from one held-out state, reporting selection regret and ranking agreement. Online latency is likewise unmeasured: candidate construction, scoring, conflict resolution, and commitment need timing apart from simulation advancement, with percentiles and 60-second overruns.

The networks consume order and yard-state records only; no layout geometry enters them, so transfer to another handling system needs no layout-specific inputs. The unvalidated part is the uncalibrated simulator, and no claim is made about specific parallel or perpendicular terminals. Cost-weight, peak-width, and longer-run robustness also remain untested by the existing frequency sweep.

6 Conclusion

Separate proposal and acceptance policies learn from counterfactual cost differences to revise spatial-temporal assignments. Across 20 independent months, time-only adjustment replicated the historical saving (cheaper in 16 months; 14/14 with a positive interval once deadlocked runs are removed), whereas the joint and block-only policies were consistently more expensive: for this trained model and simulator revision, spatial adjustment does not replicate. Operational feasibility remains unestablished.

Deployment would require finite booking quotas, carrier-approved windows, minimum notification time, and cargo/vessel deadlines in a feasibility layer [2, 3, 12], together with external waiting and rescheduling costs; Sect. 5.4 bounds the charge that would erase the saving but does not test behaviour under a binding quota. Remaining work includes repairing the crane deadlock, candidate ranking, acceptance ablations, online timing, and simulator calibration. TOS and the Busan port community system give an application context; the operator-carrier-driver workflow still needs integration and user evaluation.

Acknowledgments. This work was supported by the Public-Private Joint Technology Commercialization R&D program (Technology Transfer and Commercialization, RS-2026-25529952) funded by the Ministry of SMEs and Startups (MSS, Korea).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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